INDIVIDUAL ML PROJECT -

Project: (Example) House Price Prediction

Objective :

Develop and evaluate Machine Learning models for predicting house prices using available house-related features.

Part A: Problem Understanding

1. Define the problem.
2. Identify input variables.
3. Identify target variable.
4. Identify ML type.
5. Explain why **regression /Classification** is appropriate.

Part B: Dataset Understanding

1. Load dataset.
2. Display first five records.
3. Find dataset dimensions.
4. Identify data types.
5. Identify missing values.
6. Identify duplicate records.
7. Identify numerical and categorical features.

Part C: Data Preprocessing

1. Handle missing values.
2. Handle duplicate records.
3. Encode categorical variables.
4. Identify and handle outliers if required.
5. Apply feature scaling if required.
6. Explain every preprocessing decision.

Part D: EDA

Create at least:

- 4 visualizations
- 4 observations

Part E: Feature and Target Selection

Identify:

X = Features

y = House Price

Explain your feature selection.

Part F: Train-Test Split

Perform:

80% Training

20% Testing

and explain why.

Part G: Model Selection

Implement at least **two regression algorithms**.

Example:

1. Linear Regression
2. Decision Tree Regression
3. Explain why each was selected.

Part H: Model Training

Train both models and generate predictions.

Part I: Model Evaluation

Calculate:

- MAE
- MSE
- RMSE
- R^2

Prepare a comparison table.

Part J: Model Visualization

Create suitable visualizations such as:

- Actual vs Predicted
- Residual plot
- Regression line
- Decision tree

Part K: Model Improvement

Perform at least **two experiments**.

Record:

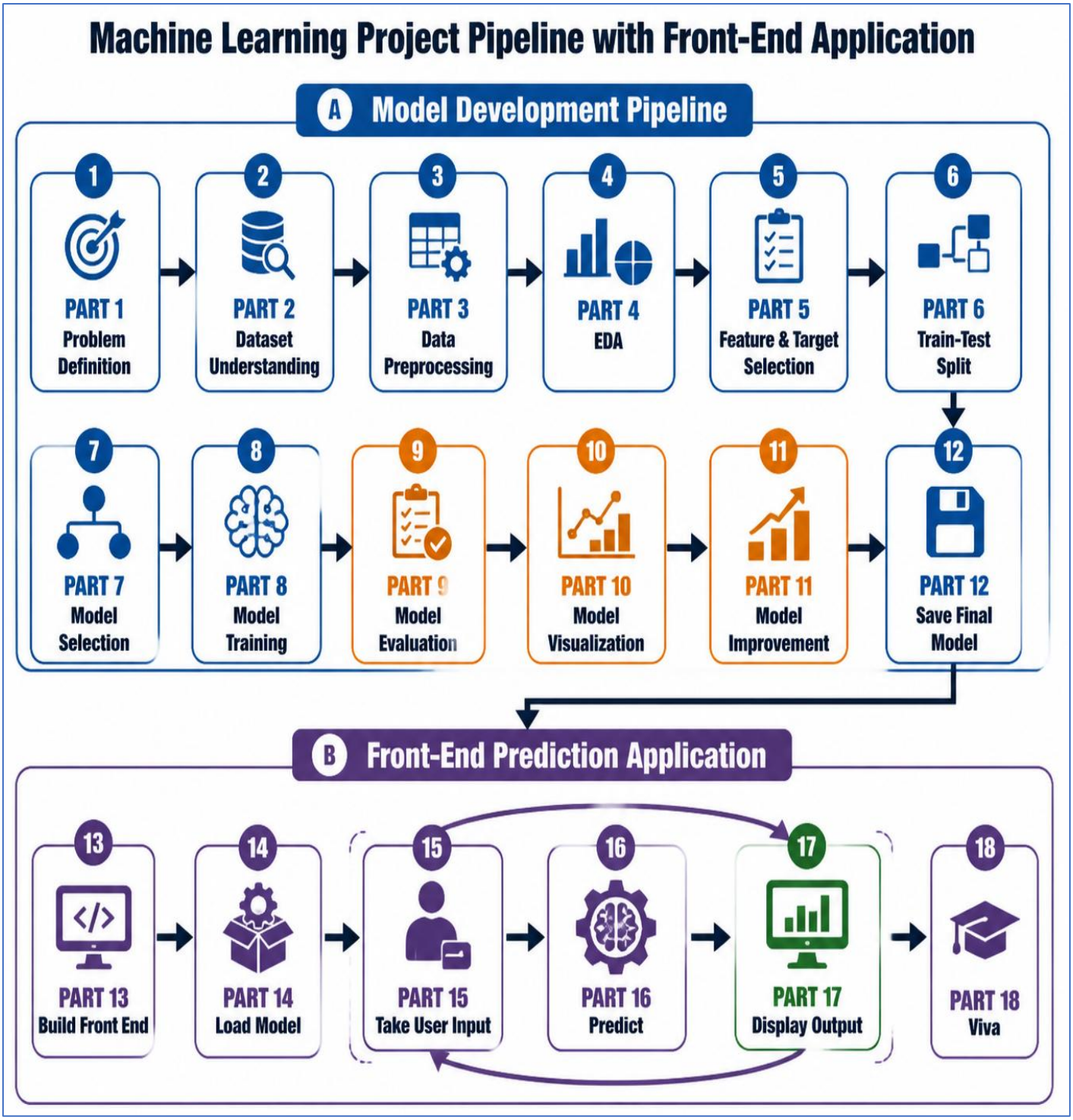
Change → Result → Observation → Conclusion

Part L: Final Interpretation

Students must answer:

1. Which model did you finally select?
2. Why?
3. What were the important features?
4. What limitations does your model have?
5. What could be improved in future?

Save Final Model > Build Front End > Load Model > Take User Input > Predict Display Output


Machine Learning Project Pipeline with Front-End Application
A Model Development Pipeline
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Viva

Project List

Roll No.	Project / Problem Statement	ML Type
1	House Price Prediction	Regression
2	Car Price Prediction	Regression
3	Used Car Price Prediction	Regression
4	Laptop Price Prediction	Regression
5	Mobile Phone Price Prediction	Regression
6	Medical Insurance Cost Prediction	Regression
7	Student Marks Prediction	Regression
8	Student Final Grade Prediction	Regression
9	Employee Salary Prediction	Regression
10	Sales Revenue Prediction	Regression
11	Advertising Sales Prediction	Regression
12	Electricity Consumption Prediction	Regression
13	Crop Yield Prediction	Regression
14	Temperature Prediction	Regression
15	Restaurant Revenue Prediction	Regression
16	Delivery Time Prediction	Regression
17	Flight Fare Prediction	Regression
18	Taxi Fare Prediction	Regression
19	Rental House Price Prediction	Regression
20	Gold Price Prediction	Regression
21	Student Performance Classification	Classification
22	Loan Approval Prediction	Classification
23	Credit Risk Classification	Classification
24	Customer Churn Prediction	Classification
25	Email Spam Detection	Classification
26	SMS Spam Detection	Classification
27	Heart Disease Classification	Classification
28	Cancer Classification	Classification
29	Diabetes Prediction	Classification
30	Iris Flower Classification	Classification
31	Wine Quality Classification	Classification
32	Mushroom Edibility Classification	Classification
33	Bank Customer Subscription Prediction	Classification
34	Employee Attrition Prediction	Classification
35	Hotel Booking Cancellation Prediction	Classification
36	Online Purchase Prediction	Classification

37	Customer Complaint Classification	Classification
38	Fraud Transaction Detection	Classification
39	Fake News Classification	Classification
40	Sentiment Classification of Reviews	Classification
41	Product Review Sentiment Analysis	Classification
42	News Category Classification	Classification
43	Handwritten Digit Classification	Classification
44	Fruit Classification	Classification
45	Animal Classification Using Features	Classification
46	Loan Default Prediction	Classification
47	Insurance Claim Prediction	Classification
48	Employee Promotion Prediction	Classification
49	Customer Purchase Prediction	Classification
50	Telecom Customer Complaint Prediction	Classification
51	Customer Segmentation	Clustering
52	Mall Customer Segmentation	Clustering
53	Retail Customer Segmentation	Clustering
54	Online Shopper Segmentation	Clustering
55	Student Grouping Based on Performance	Clustering
56	Employee Segmentation	Clustering
57	Product Grouping Based on Characteristics	Clustering
58	Movie User Segmentation	Clustering
59	E-commerce Customer Segmentation	Clustering
60	Bank Customer Segmentation	Clustering
61	Wine Dataset Dimensionality Reduction	PCA
62	Lung Cancer Dataset Dimensionality Reduction	PCA
63	Iris Dataset Dimensionality Reduction	PCA
64	Customer Dataset Dimensionality Reduction	PCA
65	Handwritten Digit Dataset Dimensionality Reduction	PCA
66	Movie Recommendation System	Recommendation
67	Book Recommendation System	Recommendation
68	Product Recommendation System	Recommendation
69	Music Recommendation System	Recommendation
70	Restaurant Recommendation System	Recommendation

Example:

Roll Number 25

Name of Student ABC

Project: SMS Spam Detection

Aim: To develop a Machine Learning model that classifies SMS messages as spam or legitimate messages.

Objective:

1. Understand text classification.
2. Preprocess SMS data.
3. Convert text into numerical features.
4. Train classification models.
5. Evaluate the classifier.
6. Compare model performance.
7. Build a simple user interface where the user enters an SMS.
8. Display **Spam / Not Spam** as the output.

Input: "Congratulations! You have won a free prize..."

Output:

Prediction: SPAM

Actual: SPAM

PROJECT 1: HOUSE PRICE PREDICTION

Aim: To develop a Machine Learning model that predicts the selling price of a house based on its characteristics.

Objective :

Students should be able to:

1. Understand a regression problem.
2. Analyze and preprocess housing data.
3. Select suitable features and target variable.
4. Train regression models.
5. Evaluate the models using MAE, MSE, RMSE and R^2.
6. Compare different regression algorithms.
7. Improve the selected model.
8. Develop a simple front end for predicting house price from user input.

Problem Statement :

Develop an ML-based House Price Prediction system that predicts the price of a house using features such as area, number of bedrooms, location-related information and other available house characteristics.

Dataset:

UCI Housing Dataset / Boston Housing-style dataset

The UCI repository contains the classic Housing dataset and provides access to a large collection of ML datasets. UC Irvine ICS

UCI Machine Learning Repository

Suggested Algorithms

- Linear Regression
- Decision Tree Regression
- Random Forest Regression

PROJECT 2: STUDENT PERFORMANCE PREDICTION

Aim:

To develop a Machine Learning model for predicting student academic performance using academic, demographic and social-related attributes.

Objectives :

Students should be able to:

1. Analyze educational data.
2. Identify important factors related to student performance.
3. Perform data preprocessing.
4. Apply regression and/or classification.
5. Evaluate the model.
6. Visualize relationships between student attributes and performance.
7. Compare ML algorithms.
8. Develop a simple prediction interface.

Problem Statement :

Develop a Machine Learning system that predicts a student's academic performance using information such as previous grades, study time, attendance-related attributes, demographic information and other available features.

The UCI Student Performance dataset contains 649 instances and 33 features and can be used for both classification and regression tasks. UCI Machine Learning Repository

Dataset :

UCI Student Performance Dataset

Suggested Algorithms

- Linear Regression
- Decision Tree
- Random Forest
- KNN

PROJECT 3: LOAN APPROVAL PREDICTION

Aim

To develop a Machine Learning classification model that predicts whether a loan application is likely to be approved based on applicant information.

Objectives

Students should be able to:

1. Understand binary classification.
2. Handle categorical and numerical data.
3. Perform missing-value treatment.
4. Encode categorical variables.
5. Train classification models.
6. Evaluate models using classification metrics.
7. Analyze the confusion matrix.
8. Build a simple loan approval prediction interface.

Problem Statement

Develop a Machine Learning system that predicts whether a loan application will be approved based on applicant attributes such as income, education, credit history, loan amount, employment status and property area.

A Kaggle loan-eligibility dataset provides applicant information and a binary Loan_Status target. Kaggle

Dataset

Kaggle Loan Eligibility Prediction Dataset

Suggested Algorithms

- Logistic Regression
- Decision Tree
- KNN
- Random Forest

PROJECT 4: CUSTOMER CHURN PREDICTION

Aim:

To develop a Machine Learning classification system that predicts whether a customer is likely to leave a service.

Objectives

Students should be able to:

1. Understand a real-world classification problem.
2. Analyze customer data.
3. Perform preprocessing.
4. Handle categorical variables.
5. Train classification models.
6. Evaluate using accuracy, precision, recall and F1-score.
7. Analyze the confusion matrix.
8. Identify important factors associated with customer churn.

Problem Statement

Develop a Machine Learning model that predicts whether a customer will continue using a service or leave the service based on customer characteristics and service usage information.

Dataset

For this project, you can use a Telco Customer Churn dataset from Kaggle.

Kaggle Datasets

Suggested Algorithms

- Logistic Regression
- KNN
- Decision Tree
- Random Forest

PROJECT 5: WINE QUALITY PREDICTION

Aim:

To develop a Machine Learning model that predicts wine quality based on physicochemical properties.

Objectives

Students should be able to:

1. Analyze numerical datasets.
2. Perform EDA.
3. Study correlations between features.
4. Apply classification or regression.
5. Compare ML models.
6. Evaluate model performance.
7. Visualize important relationships.
8. Analyze the effect of feature selection.

Problem Statement

Develop a Machine Learning model that predicts the quality of wine using physicochemical properties such as acidity, residual sugar, chlorides, alcohol and other available attributes.

The UCI Wine Quality dataset contains physicochemical input variables and wine quality as the output. It can be treated as either a classification or regression problem. UCI ICS MLR

Dataset

UCI Wine Quality Dataset

Suggested Algorithms

Option A, Classification

- Logistic Regression
- Decision Tree
- KNN

Option B, Regression

- Linear Regression
- Decision Tree Regression
- Random Forest Regression

PROJECT 6: CUSTOMER SEGMENTATION

Aim:

To segment customers into meaningful groups using unsupervised Machine Learning.

Objectives

Students should be able to:

1. Understand unsupervised learning.
2. Perform customer data analysis.
3. Select appropriate features.
4. Apply feature scaling.
5. Apply K-Means clustering.
6. Determine a suitable number of clusters.
7. Visualize clusters.
8. Interpret the resulting customer groups.

Problem Statement

Develop a Machine Learning system that groups customers into different segments based on their purchasing behavior and other available customer attributes.

Dataset

You can use the Online Retail dataset from UCI. It contains transactional data from a UK-based online retailer and has more than 500,000 transactions. UCI Machine Learning Repository

UCI Online Retail Dataset

Suggested Algorithms

- K-Means
- Hierarchical Clustering

PROJECT 7: BREAST CANCER CLASSIFICATION

Aim:

To develop a Machine Learning classification model for classifying breast tumors using extracted diagnostic features.

Objectives

Students should be able to:

1. Understand binary classification.
2. Analyze numerical medical features.
3. Preprocess the dataset.
4. Apply feature scaling.
5. Train classification models.
6. Evaluate using confusion matrix, precision, recall and F1-score.
7. Compare different algorithms.
8. Visualize model performance.

Problem Statement

Develop a Machine Learning classification system that classifies a breast tumor as benign or malignant using numerical features calculated from digitized fine-needle aspirate images.

The UCI Breast Cancer Wisconsin Diagnostic dataset contains 569 instances and 30 features describing characteristics of cell nuclei. UCI Machine Learning Repository

Dataset

UCI Breast Cancer Wisconsin Diagnostic Dataset

Suggested Algorithms

- Logistic Regression
- KNN
- Decision Tree
- SVM
- Random Forest